

## Chapter 5 Study Sheet

Congruent Triangles — everything you need on one page.

### 5.1 Angles of Triangles

- Triangle Sum Theorem: the three interior angles add to  $180^\circ$ .
- Exterior Angle Theorem: an exterior angle equals the SUM of the two remote interior angles — the two it is not adjacent to.
- Corollary: the acute angles of a right triangle are complementary (they add to  $90^\circ$ ).
- Corollary: each angle of an equiangular triangle measures  $60^\circ$ .
- Third Angles Theorem: two pairs of congruent angles force the third pair to be congruent.
- By sides: scalene · isosceles · equilateral. By angles: acute · right · obtuse · equiangular.

### 5.2 Congruent Polygons

- Congruent polygons have ALL corresponding sides and ALL corresponding angles congruent — six pairs for triangles.
- The ORDER of the letters is the information.  $\triangle ABC \cong \triangle DEF$  pairs  $A \leftrightarrow D$ ,  $B \leftrightarrow E$ ,  $C \leftrightarrow F$ . Rewriting the order changes the claim.
- Triangle congruence is Reflexive, Symmetric, and Transitive.

### Triangle Rigidity — SSS, SAS, ASA, AAS, HL

- Three well-chosen pairs are enough: SSS · SAS (included angle) · ASA (included side) · AAS (non-included side) · HL (right triangles, hypotenuse + leg).
- SSA is NOT a criterion — the second side can swing to two positions.
- AAA is NOT a criterion — same shape, possibly different size.
- Free parts: shared side (Reflexive) · vertical angles · bisected angles and segments · alternate interior angles from parallel sides.

### 5.4 Equilateral and Isosceles Triangles

- Base Angles Theorem: congruent SIDES  $\implies$  the ANGLES opposite them are congruent.
- Converse: congruent ANGLES  $\implies$  the SIDES opposite them are congruent.
- Equilateral  $\iff$  equiangular, and every angle measures  $60^\circ$ .
- Vocabulary: legs (the congruent sides) · vertex angle (between the legs) · base (opposite the vertex angle) · base angles.

### 5.7 Using Congruent Triangles — CPCTC

- CPCTC = Corresponding Parts of Congruent Triangles are Congruent.
- It is always the LAST step, never the first. Prove the triangles congruent with a criterion, THEN use CPCTC on the part you actually want.
- Two-step plan: (1) which two triangles, and which criterion? (2) which part does CPCTC give me?
- Real-world use: measure a distance you CAN reach, prove the triangles congruent, and transfer the measurement across the part you cannot reach.